

CWT-ADTH Duct Type temperature and humidity Transmitter Manual (RS485 type)



Introduction:

The transmitter is professionally used for pipeline temperature and humidity measurement. It adopts standard industrial interface RS485 output, and can be connected to on-site digital display, PLC, invert, industrial control host and other equipment. The imported temperature and humidity measuring unit has small drift and high precision. The pipeline installation method is convenient for on-site installation. It adopts anti-interference circuit design and can withstand various strong electromagnetic interferences such as on-site inverse. The equipment adopts waterproof casing design. The probe filter net uses 25um high-strength stainless steel material to ensure the entry of gas molecules. It also prevents the ingress of dust particles and water droplets, and can be applied to wet and high dust occasions for durability.

Features:

1. Temperature and humidity acquisition, RS485 output;
2. Temperature accuracy $\pm 0.3^{\circ}\text{C}$, humidity accuracy $\pm 3\% \text{ RH}$, high precision, low drift;
3. The terminal block adopts the military-grade spring-type screw-free terminal, which can be connected by one press and one plug. Even if there is no screwdriver, the line can be quickly connected, and the wire diameter can be adjusted to 0.3~2.0mm²;
4. Using special EMC anti-interference device, the site can withstand strong electromagnetic interference, industrial grade processing chip, wide range of use;
5. Support AC and DC power supply, DC 10 ~ 30V wide voltage range power supply, AC power supply voltage $24\text{V} \pm 10\%$, power input without positive and negative points, can be applied to both four-wire and three-wire connection;
6. The flange has a high-quality silicone sealing ring, which ensures the tightness between the equipment and the exhaust pipe during installation, greatly reduces the outflow of gas in the exhaust pipe, and ensures the accuracy of data measurement to the utmost extent.

Specification:

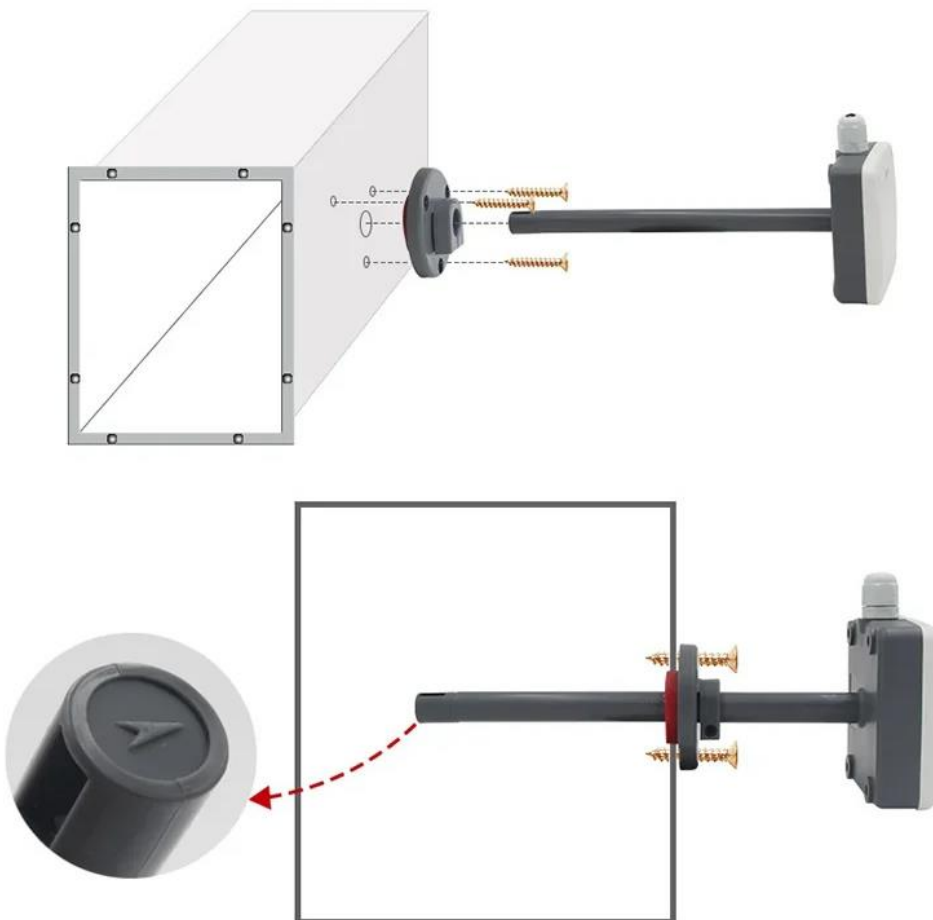
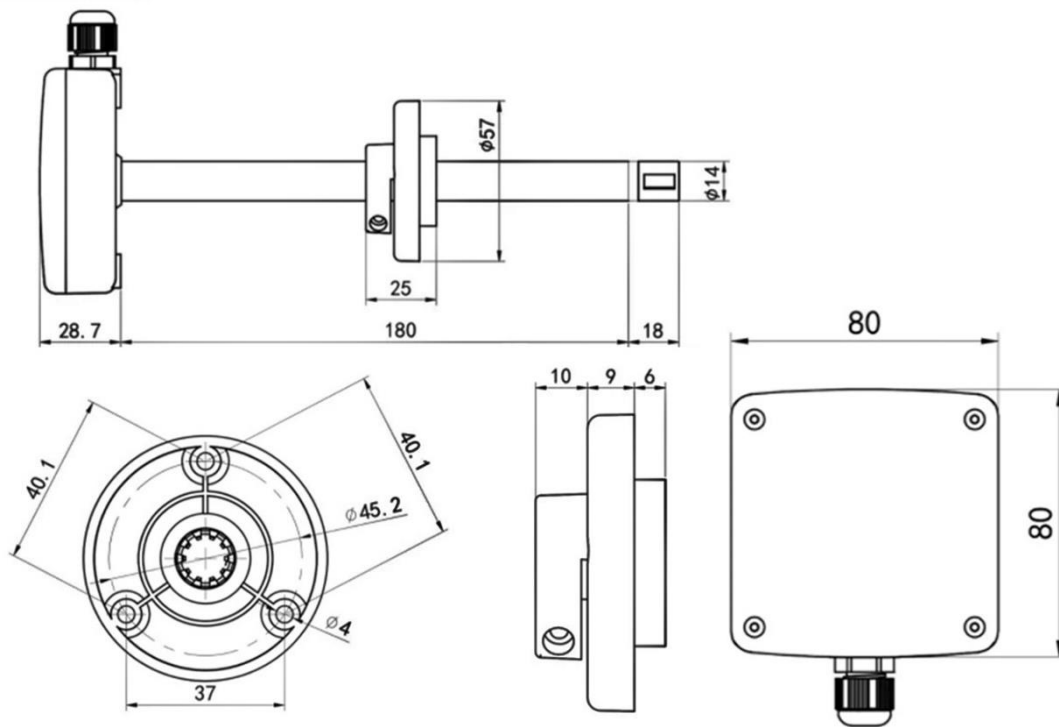
Power supply	10~30V DC	
power consumption	0.4W	
Measuring range	-40°C~+120°C, 0%RH-100%RH	
Accuracy	Temperature	±0.5°C (25°C)
	Humidity	±3%RH (60%RH,25°C)
Measure range	temperature	-40°C~+80°C
	Humidity	0%RH~100%RH
Long-term stability	Temperature	≤0.1°C/y
	Humidity	≤1%RH/y
Response time	Temperature	≤25s
	Humidity	≤8s

Wiring:



NO.	Internal identification	Description
1	A	RS485 A+
2	V+	Power+
3	GND	Power-
4	B	RS485 B-

Product size:



RS485 communication Default parameters: 4800,n,8,1

Default device address is 1

Modbus RTU protocol

Read status registers, read function code: 0x03					
Register address (Hex)	PLC Address (decimal)	meaning	Number of bytes	unit	remark
0000	40001	humidity	2	0.1%rh	Read
0001	40002	temperature	2	0.1℃	Read
Parameters registers, read function code: 0x03, write function code: 0x06					
07D0	42001	Slave ID	2	1-254	Read/Write
07D1	42002	baud rate	2	0: 2400 1: 4800 2: 9600 Default 4800	Read/Write
0050	40081	Temperature calibration value	2	0.1℃	Read/Write
0051	40082	Humidity calibration value	2	0.1%rh	Read/Write

E.g., master read temperature humidity:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x00	0x00	0x02	0xC4	0X0B

Sensor responds:

Address	Function Code	Number of byte	Humidity value	Temperature value	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x04	0x01 0xE6	0xFF 0x9F	0x1B	0xA0

Temperature calculates:

When temperature less than 0, value will be responded in complement

Temperature: FF9F H= -97 => temperature= -9.7℃

Humidity: 1E6 H= 486 => humidity= 48.6%

Set slave ID

E.g., set slave ID=2, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Set baud rate

E.g., set baud rate to 9600, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Enquiry slave ID

Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x07	0xD0	0x00	0x01	0x91	0x59

Sensor responds:

Address	Function Code	Number of Points	address	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x02	0x00 0x01	0x50	0x50